For example these rules align with the experimental design requirements for drug development studies advocated in ISO or FDA standards, but not the ICH Q2(R2) guidelines [3] that recommend a minimum number of replicates $J = 3$. The THEOPHYLLINE study is a good example of such a balanced design that can be summarized in a condensed form as follows:

$I = 6$	Number of measurement series for each material, performed over six days.
$J = 2$	Number of replicates per series.
$K = 6$	Number of validation materials with known target concentration levels, set between 0.20 and 10.0 µg/l.

We propose the concise notation $6s/2r/6l$ for this design, where s stands for the series, r for replicates, and l for levels. In this example, the choice results from a compromise between the elution time for each run and the need to establish the MAP over a very wide concentration range.

5.4.3 Check the Number of Efficient Measurements

When the experimental design is in harmony with the above-stated rules, it is possible to compute the number of effective measurements denoted N_E with formula (5.9) presented above and recalled here.

$$N_E = \frac{(A+1)^2}{\frac{\left(A+\frac{1}{J}\right)^2}{I-1} + \frac{1-\frac{1}{J}}{IJ}} \quad \text{With} \quad A = \frac{s_B^2}{s_r^2}$$

N_E, corresponds to the estimated number of df, according to the well-established Satterthwaite's approximation procedure [30]. The probability β and this number of df are used to obtain the quantile of the Student's t distribution law which appears in the formula of the coverage factor of the β-ETI (Eq. 5.10). Finally, N_E depends on three parameters: two are chosen *a priori* by the experimenter, i.e. I is the number of series, and J is the number of replicates per series; the third is the variance ratio A of the between-series variance to the repeatability variance (or within-series variance) and is only known *a posteriori* at the end of the experiment.

Being the number of df, N_E is the number of independent quantities which can be assigned to a parameter estimate and be statistically interpreted as the number of measures that efficiently contribute to the β-ETI. The higher this number, the more reliable the estimate and, in this case, the smaller the coverage factor. It is, therefore, an important indicator of what can be called to as the "quality of the study." Most importantly, in terms of scientific or financial optimality, N_E allows us to check whether the parameters of the experimental design have been correctly chosen and whether the explanation of the β-ETI intervals is satisfactory.

In addition, N_E also lets us complete the experimental design at the best cost by calculating the optimal number of runs that would eventually have to be added to obtain narrower intervals. In view of its role, it is interesting to simulate the values that N_E takes as a function of the three parameters on which it depends, I, J, and A.

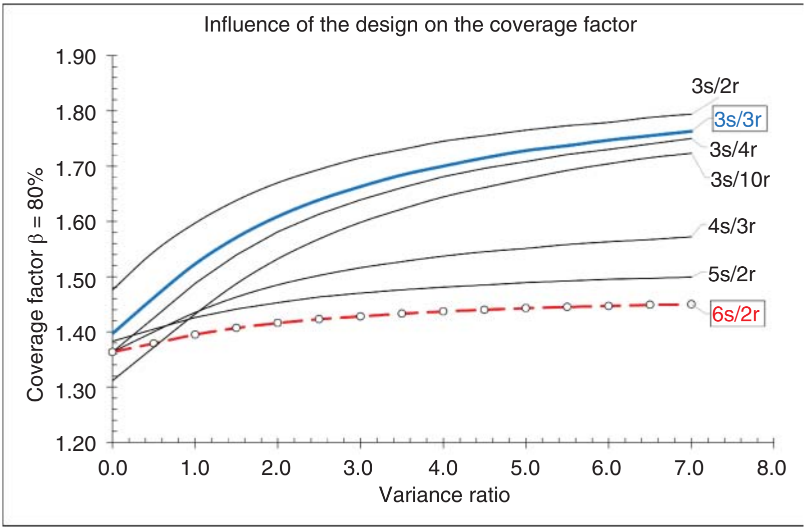

Figure 5.13 Influence of the experimental design parameters on the number of effective measurements. A design is identifiable by *s* the number of series and *r* the number of replicates per series, i.e. *Is/Jr*.

Figure 5.13 illustrates these variations as a function of the variance ratio for different combinations of I and J marked by s, the number of series and r, the number of replicates, in the form Is/Jr.

It is obvious that when the variance ratio $A \to 0$, the number of efficient measurements tends towards $(IJ - 1)$, while A plays a fundamental role and underlines significant differences between the number of trials and the number of values finally carrying information, whatever the various combinations of I and J.

In several official guides, the classically recommended minimum configuration is labeled $3r/3s$. In Figure 5.13, it appears as a thick solid line. As soon as the variance ratio $A > 3$, N_E falls very quickly from 8 to 3, meaning that almost two-thirds of the data do not participate in the estimation of TIs. It is remarkable that the addition of 1 replicate per series ($3s/4r$) to this design does not drastically improve the situation. In conclusion, when the between-series variance is high compared to the repeatability, this "official" design is unfavorable. It is possible to define the most effective design for a given budget.

For instance, if it is decided to pay for 12 analyses per level, the optimal configuration is $6s/2r$, or 6 series of 2 replicates/series, since N_E always remains higher than 5. In reverse the $2s/6r$ combination appears as the least interesting since N_E falls below 2 when A is increasing. In other words, less than 20% (2/12) of data are informative. The abacus of all other 12-trial designs, listed from bottom to top, $2s/6r$, $4s/3r$, $3s/4r$, and $6s/2r$, confirms this trend.

The conclusion is clear: for a fixed budget, the most often successful strategy is to make as many series as possible at the expense of the number of replicates. It is also possible to check the limits of variation of the coefficient Q, whose calculation

Table 5.9 Asymptotic values of the number of efficient measures for special values of the variance ratio *A*.

Variance ratio	$A \to 0$	$A = 1$	$A \to +\infty$
Number of efficient measures N_E	$N_E \to (IJ-1)$		$N_E \to I-1$
Coefficient Q	$Q \to 1$	$Q = \frac{2}{J+1}$	$Q \to \frac{1}{J}$

Table 5.10 THEOPHYLLINE – efficiency rates of the experimental design.

Concentration (μg/l)		0.05	0.1	0.5	1	2.5	10
Number of measures	IJ	12	12	12	12	12	12
Optimal number of *df*		11	11	11	11	11	11
Observed number of *df*	N_E	7.01	9.59	7.02	5.69	10.91	9.22
Efficiency rate	R_E	64%	87%	64%	52%	99%	84%
Variance Ratio	A	1.95	0.41	1.93	6.78	0.00	0.53

is provided by the formula (5.18).

$$\text{Coefficient } Q \quad Q = \frac{A+1}{J \times A + 1} \tag{5.18}$$

When A varies from 0 to $+\infty$, Table 5.9 gathers the extrema of N_E and Q as a function of some special values of A. Recall that if $A \to 0$, the between-series variance s_B^2 also tends to 0, $s_B^2 \to 0$. It means that there is no between-series effect, the intermediate precision standard deviation is equal to the repeatability standard deviation, and the conditions of quantification remain constant from one series to the next; this is the most favorable situation. The opposite conclusion is reached when $A \to +\infty$.

It is then possible to propose a method to verify whether the observed dataset will result in a satisfactory accuracy profile, by calculating the efficiency rate R_E of the experiment. It is obtained by ratioing the optimal number of efficient measurements, i.e. $(IJ-1)$ when $A \to 0$ and the number of efficient measurements N_E estimated from the data. It can be expressed as percent:

$$\text{Efficiency rate} \quad R_E = \frac{N_E}{IJ-1} \times 100 \tag{5.19}$$

Table 5.10 summarizes the various parameters calculated from the data of the THEOPHYLLINE study and extracted from Table 5.3. It confirms former remarks:

- At level 2.5 μg/l, $A = 0$ and $N_E = 10.91$ close to $(IJ-1) = 11$, study efficiency is almost 100%.
- At level 1.0 μg/l, $A = 6.78$, and $N_E = 5.69$ is close to $I-1 = 5$ and the efficiency falls to 52%.

Unfortunately, the variance ratio A is only known at the end of the experiment, whereas its influence is important. To overcome this drawback, the organization of

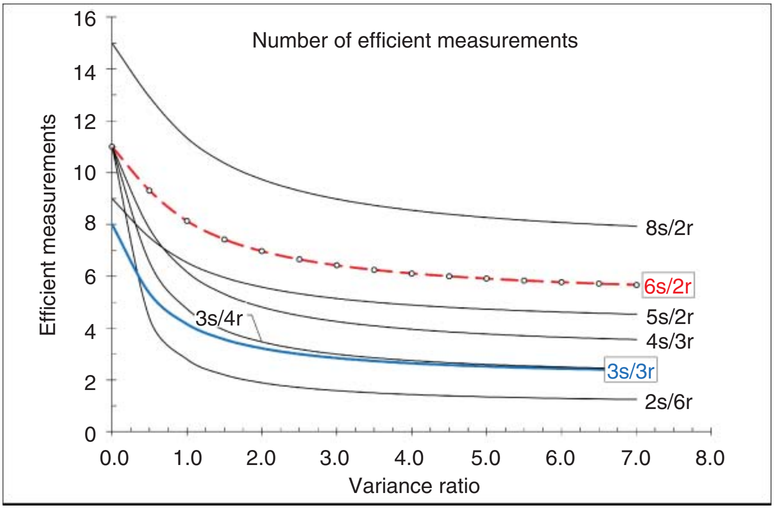

Figure 5.14 Influence of the experimental design on the coverage factor with $\beta\% = 80\%$.

the experimental design can be considered as an evolutionary process. Initially, a favorable configuration is chosen, for example, $5s/2r$, and the trials are conducted. Then, the number of series can eventually be increased according to the values of the ratio A and by taking, as an optimality criterion, the coverage factor of the interval, which is given by the following formula:

$$\text{Coverage factor (quantile of the Student's law)} \quad k_{IT} = t_{N_E, \frac{1+\beta}{2}} \tag{5.20}$$

This parameter depends on N_E, and consequently, on the A ratio. Using observed data, it is possible to predict how many additional series it would be interesting to add to the previous ones to reduce the coverage factor as much as possible and remain within the budget. The only constraint is to keep J the number of replicates/series constant so that the design remains balanced.

Figure 5.14 uses the same design labeling convention as Figure 5.13 to illustrate the consequences of the coverage factor reduction when the number of series is increased rather than the number of replicates. Once the design contains six series, the variance ratio has almost no influence. Once the variance ratio A is estimated, it is possible to simulate the consequences of adding new measurement series to the existing ones and check whether the coverage is reduced.

Such simulations using the parameters obtained at the issue of the THEOPHYLLINE study are compiled in Table 5.11. Initially, the design included six series; two were added, and the new coverage factors were computed, supposing that the variance ratio remained constant. The reduction of the β-ETI width is small, about 2–4%, and it is not interesting to add more measurements. It can be explained by the small values of the ratio A and the good experiment efficiency rates R_E, which confirms that this validation is satisfactory.

The publication on theophylline also presents the results obtained with the same method applied to caffeine quantification [23]. The experimental design $6s/2r$ is

Table 5.11 Simulation of the addition of 2 series of measurements to the initial experimental design.

Theophylline						
Concentration (μg/l)	0.05	0.1	0.5	1	2.5	10
Number of series (I)	6	6	6	6	6	6
Number of replicates (J)	2	2	2	2	2	2
Variance ratio	1.95	0.41	1.93	6.78	0.00	0.53
Initial coverage factor 80%	1.41	1.38	1.41	1.45	1.36	1.38
Added series	2	2	2	2	2	2
Updated N_E	9.79	13.30	9.81	7.96	14.93	12.81
Updated coverage factor 80%	1.37	1.35	1.37	1.40	1.34	1.35
Decrease in β-ETI width	−3%	−2%	−3%	−4%	−2%	−2%
Caffeine						
Concentration (μg/l)	0.02	0.15	0.5	1	2.5	10
Variance Ratio	25.20	1.14	1.17	6.81	0.79	0.00
Initial N_E	5.20	7.90	7.80	5.70	8.60	10.90
Initial coverage factors 80%	2.55	2.31	2.32	2.49	2.28	2.20
Updated N_E	7.3	11.0	11.0	8.0	11.9	14.9
Updated coverage factor 80%	1.41	1.36	1.36	1.40	1.36	1.34
Decrease in β-ETI	−45%	−41%	−41%	−44%	−40%	−39%

the same, but only validation material reference values vary. For theophylline, the addition of two new series is not especially useful, while for caffeine, it is much more profitable because variance ratios are much higher. In this case, the β-ETI could is reduced up to 45% at a low cost, as illustrated in the second part of Table 5.11.

The variation in the coverage factor k_{IT} as a function of the number of efficient measurements, i.e. the number of df is illustrated in Figure 5.15 for three classic values of the β. It corresponds to the quantile of the Student's t distribution law for noninteger numbers of df. As explained in Section 5.3.1 about Resource H, it can be obtained by two methods: an approximate value using the built-in Excel function called `TINV`; or an exact value using Python `t.ppf` function.

Both results are reported in Figure 5.15, with the exact value as a continuous line and the interpolate as a broken dashed line. When the number of df is below 4, the differences between both values can be significant. To summarize, some recommendations can be extracted from this figure to check if the validation study went well, but they must be balanced according to the application being treated:

- Is the number of effective measures $N_E \geq 5$?
- Is the variance ratio $A \leq 2$?

The variance ratio A can also be interpreted as a robustness parameter over time as it measures the between-series variability over the total variability. The performance of the method can be assumed as stable if A is small when shifting from one series to